12401067.

B.Tech - 1st Semester (Applied Mathematics)

Time allowed: 1hr

Max.Marks: 15

Note: Section A is compulsory and attempt any two questions from section B. Mention your group on the top of the answer sheet.

SECTION - A

1. (a) Find the rank of the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 2 & -1 & 4 \\ -2 & 8 & 2 \end{bmatrix}$

(b) If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{2}\cot u$

(c) Find df/dt at t = 0 where $f(x,y) = x\cos y + e^x \sin y,\ x = t^2 + 1,\ y = t^3 + t$

(d) Using differentials find approximate value of $\cos 44^o \sin 32^o$

(e) Show that eigen values of Hermitian matrix are real.

$\lambda = ?$

SECTION - B

2. Find the eigen values and eigen vectors of matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ ~~eigen values~~

3. Find the values of p and q for which system of equations $x + 2y + z = 6,\ x + 4y + 3z = 10,\ x + 4y + pz = q$ has (i) unique sol (ii) infinite many sol (iii) no sol

4. (a) Find relative maximum and minimum value of function $f(x,y) = 3x^2 - y^2 + x^3$.

(b) For the function $f(x,y) = \begin{cases} \frac{xy^3}{x+y^2} & ; (x,y) \neq (0,0) \\ 0\ ; & (x,y) = (0,0) \end{cases}$ find $f_{xy}(0,0)$ and $f_{yx}(0,0)$.

Basic and Applied Sciences
B.Tech-1st Semester
Applied Mathematics (BAS - 102)

12101678

Time allowed: 1hr

Max.Marks:15

Note: Section A is compulsory and attempt any two questions from section B. Mention your group on the top of the answer sheet.

SECTION - A

from kumar

1. (a) Find the rank of the matrix $\begin{bmatrix} 2 & 1 & -2 \\ -1 & -1 & 1 \\ 3 & 1 & -2 \end{bmatrix}$
   (b) Define Hermitian and Orthogonal matrix.
   (c) Prove that A and A^T has same eigenvalues.
   (d) State Cayley Hamilton Theorem.
   (e) For symmetric matrix A show that BAB^T is also symmetric.

(5 * 1 = 5)

SECTION - B

2. (a) For the function $f(x,y) = \begin{cases} \frac{x^2y(x-y)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

   Show that $f_{xy} \neq f_{yx}$ at (0,0).
   (b) Prove that eigen values of symmetric matrix are real.
3. Find eigenvalue and eigenvectors of the matrix A = $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$
4. Determine system of equations are consistent, if they are consistent find the solution 5x+3y+14z=4, y+2z=1, x+y+2z=0, 2x+y+6z=2

MST-1 (APPLIED MATHEMATICS-I)

COMPUTER SCIENCE& ENGINEERING

Time:1 hour

Max marks 15

Note: Section A is compulsory. Do any two questions from section B.

Section A (Compulsory)

1. a). Find rank of the matrix $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 4 \\ 4 & -1 & 7 \\ 2 & 1 & 5 \end{bmatrix}$

b) Prove that eigen values of Hermitian matrix are real.

c) Find approximate value of $(4.05)^{\frac{1}{2}}(7.97)^{\frac{1}{3}}$.

d) State Cayley Hamilton theorem

e) For the function $f(x,y) = \begin{cases} \frac{x^2+y^2}{x-y}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

Find $f_x(0,0)$ *and* $f_y(0,0)$. (1*5=5)

Section B

2. Solve $x + 2y - 2z = 1, 2x - 3y + z = 0, 5x + y - 5z = 1, 3x + 14y - 12z = 5.$

3. If $\theta = t^n e^{\frac{-r^2}{4t}}$, for what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$.

4. Find the eigen values and eigen vectors of A= $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (2*5=10).

Basic and Applied Sciences

B.Tech-1st Semester

Applied Mathematics (BAS - 102)

Time allowed: 1hr

Max.Marks:15

Note: Section A is compulsory and attempt any two questions from section B. Mention your group on the top of the answer sheet.

SECTION - A

1. (a) Find the integrating factor of $(x^3 + y^3 + 1)dx + xy\,dy = 0$

   (b) Solve the differential equation $xy' + 3y = x^3y^2$

   (c) Define unit step function at point p.

   (d) Find the inverse of Laplace Transform of $\frac{s(1+e^{-s\pi/2})}{s^2+4}$

   (e) State and prove first shifting theorem.

(5 * 1 = 5)

SECTION - B

2. Find the general solution of equation $x^2y'' - 2xy' - 4y = 6x^2 + 4\ln x$

3. (a) Solve by variation of parameter $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$

   (b) Form the differential equation whose particular solution is $1 + x + e^x - 3e^{3x}$.

4. Solve the initial value problem $y'' - 6y' + 5y = e^{2t}$ such that $y(0) = 1, y'(0) = -1$ using Laplace transformation.